

Amazon Marketplace Sales Analysis

Business problems, insights and recommendations

Data period: 1 January 2020 to 30 July 2024

Scope

21,629 orders | 898 customers | 54 sellers | 765 products | 6 categories

Data prepared in MySQL; analysed and visualised in Power BI

Prepared October 2026

Executive summary

This report analyses more than four and a half years of marketplace activity to understand where revenue comes from, where it is lost, and which risks deserve management attention. The business is profitable and its customers are loyal, but it is exposed in three ways: sales of its dominant category have almost stopped, a very small group of customers drives all returns, and revenue depends heavily on one category, two states and a limited number of customers and sellers.

\$12.64M Gross revenue	75.4% Gross margin	21,629 Orders	13.1% Orders returned
\$10.38M Net revenue (completed orders)	\$1.27M Profit lost to returns	94.5% Buyers who reorder	89.7% Revenue from electronics

What leadership should know

- **Electronics has nearly stopped selling.** March to July orders fell from 1,443 in 2023 to 10 in 2024. Electronics is almost 90% of revenue, so this is the single largest risk.
- **Thirty customers cause every return.** They send back about 98% of what they order, costing \$1.67M in revenue and \$1.27M in gross profit.
- **Carrier return rates are misleading.** BlueDart (79%) and DHL (21.5%) look poor only because the same 30 customers use them. Other customers had no returns on either carrier.
- **Revenue is concentrated.** Ohio provides 54% and Texas 22% of revenue; 213 loyal customers provide 91.7%; the top five sellers provide 65.5%.
- **Demand is seasonal.** March to September carries roughly 1.7 times the order volume of October to February, which stock, promotions and logistics should plan around.

How this report is organised

Section	Content
1 Project overview and approach	How the data was collected, loaded into MySQL, checked, turned into KPIs and visualised in Power BI
2 Business problems and questions	The ten questions the analysis set out to answer
3 Key findings and insights	Seven findings with evidence, likely cause and action
4 Recommendations and roadmap	Prioritised actions with owners and timing
5 Limitations and next steps	What the data cannot tell us yet and what to do next

1 Project overview and approach

Objective

Build a trustworthy view of marketplace performance and use it to answer practical business questions: how sales are trending, how much revenue is lost after the order is placed, who the business depends on, and what operations should fix first. The work is intended for management, category, customer-experience, logistics and supply-chain teams.

Workflow



Step 1: Collecting and organising the data

Raw files covering orders, order items, payments, shipping, customers, sellers, products, categories and inventory were gathered and organised. File and column names were made consistent, and the grain of each table (for example one row per order line) and the keys that connect the tables were agreed before anything was loaded.

Step 2: Creating the MySQL database and importing the data

A local MySQL database was created and all nine tables were imported into it. MySQL served as the single, queryable source of truth, so every later check and figure could be reproduced.

Table	What it holds	Rows
orders	One record per order: date, customer, seller and order status	21,629
order_items	Product, quantity and price per unit for each order (one line per order)	21,629
payments	Payment status and payment date for each order	21,629
shipping	Carrier, shipping date, delivery status and return date	21,141
customers	Customer profile and state	898
sellers	Seller name and country of origin	54
products	Product name, list price and unit cost	765
category	Product category names	6
inventory	Current stock by product and warehouse	765

Step 3: Table design, keys and relationships

Each table was given appropriate data types, a primary key and the foreign keys needed to join it to the others. Orders link to customers and sellers; order items, payments and shipping link to orders; order items link to products through inventory; products link to categories. Because each order contains exactly one line item, the order-level tables relate one to one.

Step 4: Data quality checks

Before any analysis, the data was tested for duplicates, broken links, inconsistent statuses and illogical values. The outcomes below determined which numbers could be trusted and which limits to state.

Check	Outcome
Duplicate order keys	None. 21,629 unique orders, each with exactly one line item.
Price consistency	The price on every order line equals the product list price.
Orders without a shipment	488 orders have no shipment, the same number as cancelled orders.
Status consistency across tables	Failed payments (488) equal cancelled orders; refunded payments (2,840) equal returned orders.
Return dates	Populated only for the 2,840 returned shipments. Stored as text, so converted to dates for analysis.
Labels and spelling	The status "Payment Succeeded" is misspelled at source; category names use inconsistent capitalisation. Both were kept consistent in reporting.
Coverage gaps	212 of 898 customers never ordered; 15 of 765 products never sold.
Logical timing	Orders ship in about 3 days on average; returns arrive about 10 days after shipping; payments are recorded the same day as the order.
Date window	1 January 2020 to 30 July 2024, so 2024 is a partial year.

Step 5: Identifying the KPIs

With a clean base, KPIs were defined for each area of the business so that every business question has a number to test it against. The areas and their headline results are below.

Business area	What it tells us	Headline result
Revenue and profit	Size and quality of sales	\$12.64M gross revenue, 75.4% gross margin, \$584 average order
Order health	How many orders succeed	82.3% completed, 13.1% returned, 2.3% cancelled
Customers	Loyalty and dependence	686 active of 898, 94.5% repeat, 213 loyal customers
Logistics and payments	Delivery and payment reliability	About 3 days to ship, 2.3% payment failure
Sellers and products	Supplier and catalogue risk	Top 5 sellers 65.5% of revenue, 26.8% from international sellers
Inventory	Stock readiness	39,491 units, 51 low-stock SKUs, 15 unsold products

Step 6: Moving to Power BI

The cleaned data and KPIs were moved into Power BI, where an eight-page dashboard turns the numbers into something decision-makers can explore with filters. Each page is tied to a group of business questions.

Dashboard page	Purpose
1 Executive overview	One-screen health check of revenue, profit, order mix and key risks
2 Sales and growth	Yearly growth, seasonality and category trends
3 Returns and leakage	Where revenue and profit are lost and who is responsible
4 Logistics and payments	Carrier performance, shipping speed, payment failures and refunds

Dashboard page	Purpose
5 Customers	Loyalty, activation, geography and top customers
6 Sellers and products	Seller and product concentration, category margins
7 Inventory	Stock levels, low-stock and unsold products
8 Insights summary	The findings and actions in this report

Step 7: Turning analysis into insight

Each dashboard view was read against the business question behind it. Where a pattern looked like a cause, it was tested before being reported: for example, the carrier return rates were re-cut by customer group, which showed the carrier was not the real cause. Findings that rest on judgement rather than proof are labelled as likely causes to be validated.

2 Business problems and questions

The analysis was organised around ten questions, grouped by the decision they support. Section 3 answers each one in turn.

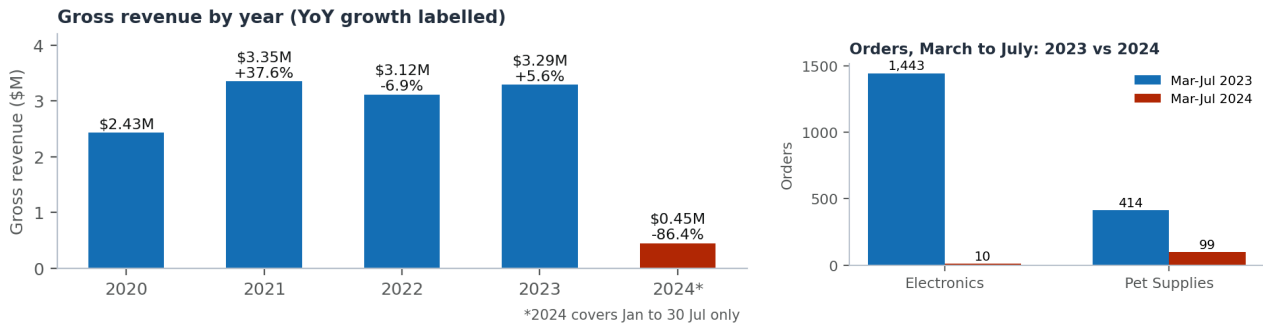
#	Theme	Business problem	Question
Q1	Growth	Revenue dropped sharply in 2024 and the main category went quiet	Why did electronics orders collapse from March 2024, and is the cause demand, supply or data?
Q2	Growth	Sales swing through the year, but plans may not reflect it	How large is the March to September peak, and are stock and promotions planned around it?
Q3	Returns	Returns remove revenue and profit after the sale	Who drives returns, and what do they cost?
Q4	Returns	Some carriers show very high return rates	Is the return problem caused by the carriers or by something else?
Q5	Returns	Returns might come from specific products or sellers	Do categories or sellers differ meaningfully in return rate?
Q6	Customers	Revenue may depend on a small group of buyers	How dependent is the business on its loyal customers?
Q7	Customers	A large share of registered customers never buy	How can the 212 customers who never ordered be activated?
Q8	Risk	Revenue may be concentrated in too few places	How exposed is the business to one category, two states and a few sellers?
Q9	Operations	Failed payments, cancellations and delays cost sales	Are payments, shipping and fulfilment working reliably?
Q10	Inventory	Stock may be misaligned with demand	Which products should be restocked first, and which never sell?

3 Key findings and insights

Each finding states what the data shows, the most probable cause, and the action it points to. Where a cause is a hypothesis rather than a proven fact, it is stated as such.

Finding 1: Electronics demand collapsed from March 2024

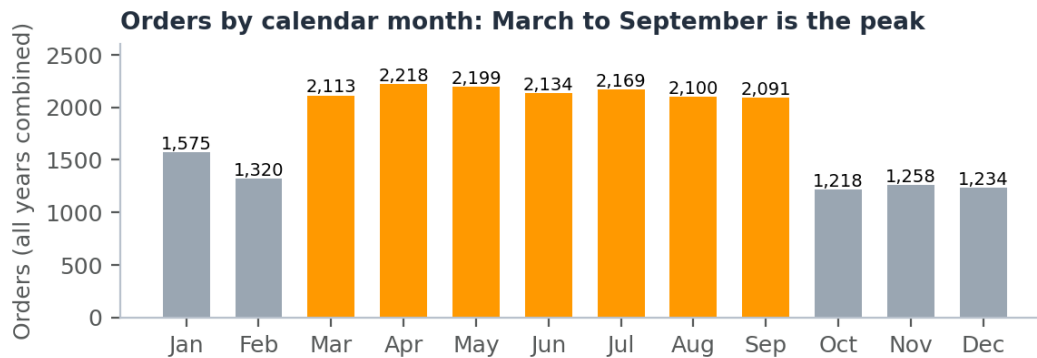
Answers: Q1



What the data shows	Gross revenue was \$3.29M in 2023 and \$0.45M in 2024 (to July). Electronics orders for March to July fell from 1,443 to 10, a 99% drop. Other categories slowed too (Pet Supplies fell from 414 to 99 orders), but none came close to this. Average order value dropped from about \$550 to about \$100 because high-priced electronics stopped appearing.
Why it is happening	The fall is concentrated in one category and begins abruptly, which points to a supply, listing, pricing or seller-availability problem or a gap in data capture, not a gradual loss of demand. This is a likely cause to confirm, not a proven one.
Business impact	Electronics is 89.7% of revenue, so the whole business is carried by the category that has stopped selling.
Recommended action	First rule out a data or catalogue issue for electronics from March 2024 (feeds, listing status, stock). If demand is genuinely lower, review pricing, competitor activity and seller availability. Track orders by category by month every week so a break like this is seen within days.

Finding 2: Demand is seasonal

Answers: Q2



What the data shows	Across all years, March to September receives roughly 2,100 to 2,200 orders per calendar month against 1,200 to 1,575 from October to February, about 1.7 times as much. In 2023, monthly orders fell from 604 in September to 333 in October.
Why it is happening	A repeating annual cycle in demand. The drop begins sharply in October and recovers in March.
Recommended action	Plan stock, carrier capacity and promotions around the March to September peak. Use targeted offers and bundles to lift the October to February trough rather than waiting for it to recover.

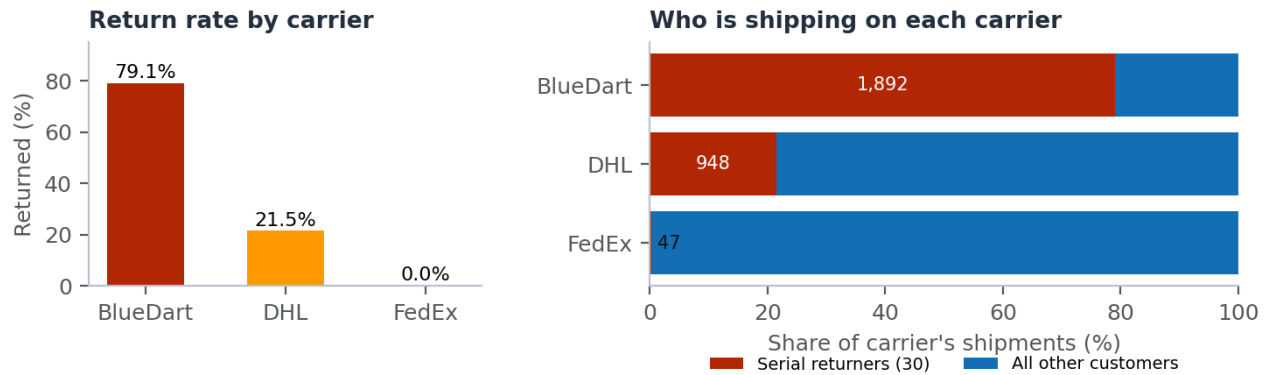
Finding 3: Thirty customers cause every return

Answers: Q3, Q5

What the data shows	30 of 686 active customers (4.4%) account for all 2,840 returned orders, with 65 to 117 returns each and a return rate of 98.4% on their own orders. Returns cost \$1.67M in revenue, 5,352 units and \$1.27M in gross profit. Return rates barely differ by category (11.6% to 14.2%) or seller (11.3% to 14.8%).
Why it is happening	Because category and seller make little difference, the cause is customer behaviour, not product quality or supplier performance. This pattern is consistent with return abuse or repeated purchase-and-return behaviour and should be investigated before any conclusion is drawn.
Business impact	13.1% of orders are returned, removing revenue that was already counted as sales.
Recommended action	Review the 30 accounts for abuse or fraud. Introduce return limits, restocking fees or manual review for repeat returners, and monitor the number of customers with returns each month.

Finding 4: Carrier return rates are a customer effect, not a carrier effect

Answers: Q4



What the data shows

Return rates are BlueDart 79.1%, DHL 21.5% and FedEx 0%. However, the 30 returners account for 1,892 of BlueDart's 2,392 shipments and 948 of DHL's 4,403, and every one of those parcels was returned. BlueDart carried 500 parcels for all other customers with no returns. Shipping time is almost identical across carriers (about 3 days).

Why it is happening

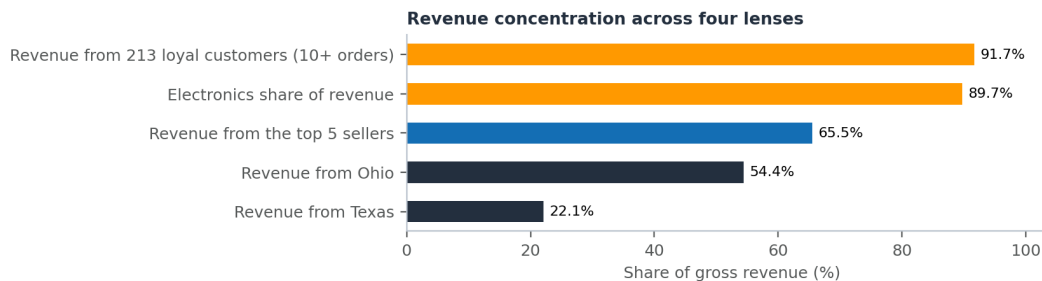
The share of each carrier's shipments that belongs to the 30 customers matches its return rate almost exactly. The carriers did not cause the returns; the customers chose them.

Recommended action

Do not drop or penalise BlueDart or DHL on return rate alone. Resolve the customer cohort first, then re-test carrier performance. Add delivery-damage and delay data, which this dataset does not hold, to judge carriers properly.

Finding 5: Revenue is concentrated in a few places

Answers: Q6, Q7, Q8



What the data shows

Electronics is 89.7% of revenue. Ohio generates \$6.87M (54.4%) and Texas \$2.80M (22.1%), while the third state, New York, generates \$0.32M. 213 customers with 10 or more orders produce 91.7% of revenue, and 94.5% of buyers reorder. The top five sellers produce 65.5% of revenue; international sellers produce 26.8%. By product the spread is healthy: the top ten products are only 17.9% of revenue. 212 customers have never ordered.

Why it is happening

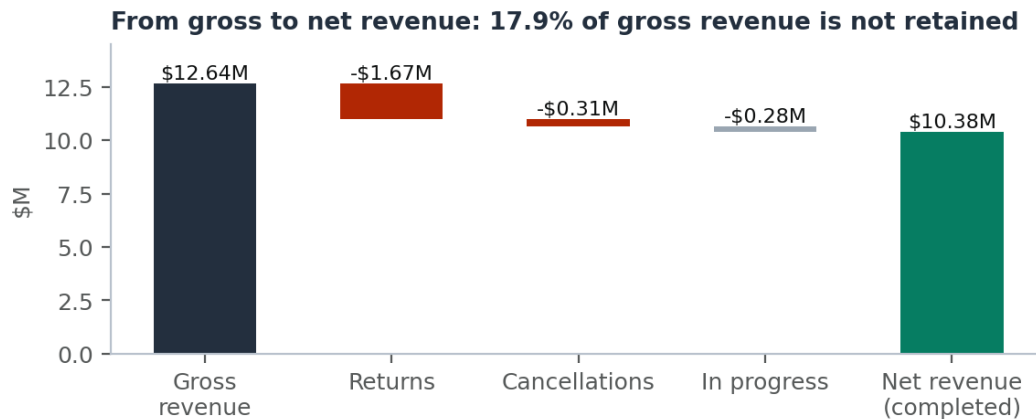
A highly loyal, repeat-buying base concentrated in two states and one category. Loyalty is a strength, but it also means losing a small number of customers or one category would hit revenue hard.

Recommended action

Set diversification targets for non-electronics categories and for states outside Ohio and Texas. Build a retention programme for the 213 loyal customers, and a reactivation campaign for the 212 customers who have never ordered.

Finding 6: About one sixth of revenue is lost after the order is placed

Answers: Q9



What the data shows	Of \$12.64M gross revenue, \$1.67M is lost to returns and \$0.31M to cancellations (15.7% in total), and a further \$0.28M is still in transit, leaving \$10.38M in net revenue. Failed payments (488) equal cancelled orders, and refunds (2,840) equal returns, so payment problems and returns explain the leakage. Orders ship in about 3 days and returns arrive about 10 days after shipping.
Why it is happening	Returns are the dominant leak (84% of the loss) and are driven by the group in Finding 3. Cancellations come from failed payments and are small by comparison (2.3% of orders).
Recommended action	Treat the returner cohort as the first lever. Then reduce payment failures with retry flows and clearer error messages, since each failed payment becomes a cancelled order.

Finding 7: Inventory is healthy overall, with a short list to act on

Answers: Q10

What the data shows	39,491 units are in stock, worth \$3.78M at cost and \$15.51M at retail, a potential gross profit of \$11.73M. 51 SKUs (6.7%) hold fewer than 10 units, and 15 products have never sold (\$2,979 of stock at cost). All stock sits in a single warehouse.
Why it is happening	Low-stock SKUs are a replenishment gap; the unsold products are a catalogue-management issue. Neither is large in value.
Recommended action	Restock the 51 low-stock SKUs in order of revenue contribution and decide whether to retire or relaunch the 15 unsold products. Treat the single-warehouse setup as a resilience risk to review.

4 Recommendations and roadmap

Actions are ordered by the value at stake and the speed with which they can be started.

Priority	Action	Owner	Timing	Value at stake
1	Investigate and resolve the electronics decline (data, listing, supply, pricing)	Category and supply teams	0 to 30 days	About 90% of revenue
1	Review the 30 serial-returner accounts for abuse or fraud	Trust and safety	0 to 30 days	\$1.67M revenue, \$1.27M profit
2	Introduce return limits, fees or review rules for repeat returners	Customer experience and policy	30 to 60 days	Prevents repeat of leakage
2	Re-test carrier performance after the cohort is handled; collect damage and delay data	Logistics	30 to 90 days	Fair carrier decisions
2	Retention programme for the 213 loyal customers	CRM and marketing	30 to 90 days	91.7% of revenue
3	Seasonal plan for stock, carriers and promotions	Planning	60 to 90 days	Peak and trough demand
3	Diversify beyond electronics and beyond Ohio and Texas; reactivate 212 inactive customers	Growth and marketing	90 days and beyond	Lower concentration risk
3	Restock 51 low-stock SKUs; retire or relaunch 15 unsold products	Supply chain	Within 30 days	Availability on best sellers

How the questions map to the findings

Question	Answered in
Q1 Electronics collapse	Finding 1
Q2 Seasonality	Finding 2
Q3 Who drives returns	Finding 3
Q4 Carrier or customer	Finding 4
Q5 Category and seller returns	Finding 3
Q6 Loyalty dependence	Finding 5
Q7 Never-ordered customers	Finding 5
Q8 Concentration risk	Finding 5
Q9 Payments and fulfilment	Finding 6
Q10 Stock priorities	Finding 7

5 Limitations and next steps

Limitations

- 2024 is a partial year (to 30 July), so its year-over-year change is not like-for-like. The electronics drop, however, is far larger than a short year can explain.
- The 75% gross margin is unusually high for retail. The unit cost field should be confirmed as a full cost before margin is used for pricing decisions.
- Inventory is a single snapshot with no history, so stock cover can only compare categories and cannot forecast stock-outs.
- The return pattern (30 customers returning about 98% of orders) is unusually clean and may partly reflect how the dataset was produced. Validate it against live data before acting.
- The dataset holds no delivery-damage, delay, marketing or customer-service data, so causes outside sales, payments, shipping and stock cannot be tested.

Next steps

- Confirm the cause of the electronics decline with category and supply teams, and refresh the data through the latest month.
- Score customers for return risk and add it to the dashboard for ongoing monitoring.
- Build a demand forecast that accounts for seasonality, and test the retention and reactivation campaigns on a small group before scaling.